

SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 2.2: Plant Transport — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 10	
Sub-Strand	Sub-Strand 2.2: Plant Transport
Driving Question	DRIVING QUESTION: How does water and food travel through an entire plant — from the soil to the highest leaf — and why does this journey sometimes fail, causing crops like maize in Kibwezi to wilt and die?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Wilting Maize of Kibwezi: Encountering the Phenomenon	We observed photographs of wilted maize crops in Kibwezi, Makueni County, where the soil was visibly damp from morning watering but the plants were still drooping by midday. We noticed that maize near the dry seasonal riverbed wilted sooner, while sukuma wiki growing in the shade of a mango tree stayed upright and healthy all day.	We learned that our first instinct — that wilting means the soil has no water — is challenged by this scenario. The soil had water, yet the plants still wilted. This tells us that there must be something happening inside the plant that we do not yet fully understand. We also learned that shade, slope, and soil moisture all seem to affect how badly a plant wilts.	We cannot fully explain the wilting yet. We have recorded our first predictions and initial ideas, but we know these may be incomplete or incorrect. Our Driving Question Board shows the questions we need to investigate. By the end of this unit, we aim to construct a complete scientific explanation of what the Kibwezi farmer is observing — one that is supported by evidence from our investigations.
Lesson 2 What Do Plants Need to Survive? — Mapping the Case for a Transport System	Students observed (via annotated plant diagrams and resource cards) that different plant resources — water, mineral salts, carbon dioxide, and sunlight — enter the maize plant at different locations: water and mineral salts at the roots below the soil, carbon dioxide and sunlight at the leaves above the soil.	Students learned that because plant resources enter at one location but are needed and used at a different location (e.g., water enters at the root but is needed at the leaf), a plant must have an internal transport system to move resources from entry point to use point — and that without this system, the leaf cannot get the water it needs and wilts, exactly as seen in the Kibwezi maize field.	Students can now begin to explain part of the Kibwezi wilting mystery: the maize plant wilts when water cannot reach the leaf fast enough through its transport system, even if water is present in the soil. The farmer's observation that 'the soil has water but the plant still wilts' makes sense because soil water and leaf water are separated by the entire length of the plant — and something must carry that water across that distance.
Lesson 3 The Big Picture: Overview of the Plant Transport System	Students observed a whole-plant diagram of a maize plant (rooted in Kibwezi soil) showing two distinct vascular systems — xylem and phloem — running continuously	Students learned that plants have two specialised vascular transport tissues: (1) Xylem — transports water and dissolved mineral salts from roots upward to leaves;	Students used their new knowledge of xylem and phloem to deepen their explanation of the Kibwezi wilting maize phenomenon: the xylem highway must

(Xylem and Phloem)	from root tip to leaf tip and to the developing maize cob. They observed that blue (xylem) arrows point strictly upward, while red (phloem) arrows are bidirectional — primarily downward but also toward the cob as a sink. They observed from the reading that xylem is composed of dead, hollow cells requiring no living machinery to function, while phloem is made of living cells.	composed of dead, lignified cells that form continuous hollow tubes; transport is unidirectional (upward). (2) Phloem — transports manufactured food substances (sugars/assimilates) from leaves to all other parts of the plant including roots and storage organs (e.g., maize cob); composed of living sieve tube cells; transport is bidirectional (source to sink). Students learned to locate both tissues within root, stem, and leaf cross-sections and to represent their directional flow using a standardised colour-coding system.	continuously deliver water from the soil (absorbed by root hairs via osmosis, as established in Lesson 2) upward through the stem to the leaves. If this upward delivery is insufficient — whether due to high evaporative demand on a hot afternoon, a long path length in tall maize, or competition from the dry riverbed soil — the leaf cells lose turgor pressure and the plant wilts. This explains why the farmer's morning watering does not guarantee afternoon hydration of the topmost leaves. Students recorded this causal chain in Model v.1 and posted evidence claims on the Driving Question Board, advancing the class's collective answer to the driving question.
Lesson 4 The Entry Point: Root Structure and Water Absorption	Students observed real maize/bean seedling roots under hand lenses and (where available) a microscope, producing labelled scientific drawings of root hair cells, cortex, endodermis, and xylem. They also observed potato strips in tap water vs. concentrated salt solution, recording changes in length/firmness after 15 minutes.	Students learned that water enters the root by osmosis — moving passively from soil (high water potential) through root hair cells, cortex, and endodermis into the xylem (low water potential). Mineral salts enter by active transport against their concentration gradient, requiring ATP. The Casparian strip in the endodermis forces all water through living cells, acting as a selective checkpoint. Root hairs increase surface area to maximise absorption rate.	Students explained that the Kibwezi farmer's maize wilts despite soil water being present because, in dry or saline soil, the water potential of the soil drops below that of the root hair cells — reversing or halting osmosis so water cannot enter (or even leaves the root). Plants near the dry riverbed wilt sooner because sandy, drier, more saline soils near the riverbed reach this critical low water potential threshold faster than the wetter lower slopes.
Lesson 5 The Highway: Stem Structure and Vascular Bundles	Students cut cross-sections of fresh bean or sunflower stems, stained them with eosin or iodine, and observed distinct coloured patches arranged in a ring pattern (in dicot stems). They noted that some regions had thicker walls and appeared more densely stained than others. They also set up split celery stalks in two different food-dye colours and observed the initial uptake of colour beginning at the cut base.	The stem contains vascular bundles made up of two distinct tissues: xylem (inner, thick-walled, dead cells that form hollow tubes for carrying water and minerals upward) and phloem (outer, thin-walled, living cells that carry dissolved sugars mainly downward from leaves). In herbaceous dicot stems the bundles are arranged in a ring; in monocot stems (like maize) they are scattered. These bundles are the physical 'highway' that connects roots to leaves.	Water and mineral salts absorbed from the soil by root hair cells travel upward through the xylem vessels in vascular bundles running through the stem to every leaf. Sugars manufactured in leaves by photosynthesis travel downward through the phloem to roots, growing tips, and storage organs. In the Kibwezi maize, extreme drought stress can cause air embolisms (cavitation) in the narrow xylem vessels, permanently breaking the water highway — which is why some plants never recover even after watering returns. Plants like the sukuma wiki shaded by the mango tree lose less water through transpiration, reducing the tension in their xylem and making cavitation far less likely.

Lesson 6 The Destination: Leaf Structure and Transpiration	Students observed water droplets inside a polythene bag tied around a leafy branch (sweating leaf bag demonstration), confirming that leaves release water vapour. They recorded results from the Lesson 5 coloured dye experiment, observing stained xylem tissue in stem cross-sections. They prepared stomatal impressions from sukuma wiki leaves using clear nail varnish and observed guard cells and stomatal pores under a hand lens, finding more stomata on the lower epidermis. They made cross-sections of the leaf midrib and identified the vascular bundle, epidermis layers, mesophyll, and waxy cuticle.	Students learned that transpiration is the loss of water vapour from a plant, mainly through stomata in the lower epidermis of leaves. They learned that xylem vessels carry water from the roots all the way to the leaf mesophyll cells, where it evaporates and exits through stomata as water vapour. They identified four factors that increase the rate of transpiration: high temperature, high wind speed, low humidity, and high light intensity — all of which are present in Kibwezi on a dry afternoon. They explained why the farmer's maize wilts: transpiration rate in the hot, windy, dry afternoon exceeds the rate at which roots can absorb water from the soil, causing a water deficit inside the plant.	Students can now explain the complete water journey from soil to atmosphere: water enters root hairs by osmosis, travels through root cortex cells, enters xylem vessels, is pulled up the stem by the transpiration pull (tension created as water evaporates from leaves), reaches leaf mesophyll cells through vein xylem, evaporates into air spaces, and exits through stomata as water vapour. The wilting of the Kibwezi maize is explained by a rate mismatch: on hot, dry, windy afternoons, stomata are open for photosynthesis, and the four transpiration factors combine to drive water loss far faster than roots can replace it, causing cells to lose turgor and the plant to wilt. The sukuma wiki under the mango tree barely wilts because shade reduces both light intensity and temperature — two of the four factors — dramatically slowing transpiration rate.
Lesson 7 The Engine: Transpiration Pull, Cohesion-Tension, and Root Pressure	Students observed an air bubble moving through a calibrated capillary tube in an improvised potometer, measuring how fast a cut leafy stem took up water under four conditions: control (still air), wind (fan/blowing), heat (lamp), and blocked stomata (petroleum jelly). They observed that wind and heat conditions produced the fastest bubble movement (highest water uptake rates), while blocking the stomata dramatically slowed the bubble. Students also observed class-pooled data posted on the board and drew bar graphs comparing uptake rates across all four conditions.	Students learned that water moves up the xylem primarily through the cohesion-tension mechanism: transpiration at the leaf surface lowers water potential, creating tension that pulls the cohesive water column upward from roots to leaves. Root pressure provides a supplementary upward push. The rate of water uptake increases when transpiration is accelerated by wind and heat — conditions that match a hot, dry Kibwezi afternoon. Blocking stomata (petroleum jelly) sharply reduces water uptake, confirming that transpiration through stomata is the primary driver of the entire upward journey.	Students explained, using the CER framework and their potometer data, that the Kibwezi maize wilts even after watering because the rate of transpiration in the hot, windy afternoon exceeds the rate at which water can be absorbed from the soil and transported up the xylem. When transpiration rate is very high, the tension in the xylem increases; if water in the soil is insufficient or absorption is too slow, the cohesive water column cannot be replenished fast enough, causing cells to lose turgor and the plant to wilt. Students updated their cumulative transport model with cohesion-tension arrows, a transpiration label at the leaf, and a root pressure label at the root.
Lesson 8 The Return Journey: Translocation and Final Model Build	Students observed annotated photographs and concentration data from girdling experiments (eucalyptus trees near Kericho; bean plant sucrose data) showing accumulation of sugars above the girdle and starvation below, providing direct evidence that phloem carries sucrose	Students learned that phloem sieve tubes, assisted by companion cells, transport sucrose and amino acids from photosynthetic source cells (leaves) to metabolic sink cells (roots, fruits, growing tips) by mass flow driven by a hydrostatic pressure gradient — the pressure-flow	Students explained the anchoring phenomenon — 'The Wilting Maize of Kibwezi' — in full: (1) Despite watered soil, the rate of transpiration through stomata on a hot, dry Kibwezi afternoon exceeds the rate of water uptake and xylem transport, creating a water deficit in leaf cells, causing

	downward from source to sink. Students also observed a physical pressure-flow demonstration using two beakers, concentrated and dilute sucrose solutions, semi-permeable membranes, and water — watching mass flow driven by osmotic pressure gradients.	(mass flow) hypothesis. Students consolidated the full plant transport system: osmosis drives water into root hairs down a water potential gradient → cohesion-tension pulls water up xylem → transpiration at stomata drives the pull → sucrose is loaded into phloem at source → pressure-flow moves sucrose to sinks → osmosis unloads water at sink. Students applied this integrated understanding to construct a complete annotated model and to explain why the Kibwezi maize wilts, why some plants die, and why shaded sukuma wiki is protected.	loss of turgor and wilting. (2) Plants near the dry seasonal riverbed wilt sooner because soil water potential is lower and root uptake is less efficient, meaning the xylem system cannot replace transpired water fast enough; if guard cells lose turgor and fail to close stomata, or if the phloem system is compromised by drought stress, the plant cannot recover and dies. (3) Sukuma wiki under the mango tree experiences lower transpiration demand because shade reduces leaf temperature and the vapour pressure gradient between leaf and air — stomata need not open as wide, less water is lost, the xylem supply is sufficient, turgor is maintained, and the plant does not wilt.
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